Enterprise Active Directory, Windows Security Monitoring & Splunk SIEM Implementation

Security Operations Lab Portfolio (LAB-01 to LAB-10)

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Role:

Security Operations (SOC) Analyst Portfolio | Security Engineering Lab Project

Project Duration

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Executive Summary

This portfolio documents the successful design, deployment, configuration, and monitoring of a complete enterprise-style Windows security environment. The project was developed to gain practical experience in Active Directory administration, Windows security controls, event collection, centralized logging, security monitoring, and Security Information and Event Management (SIEM).

The environment simulates a real-world corporate network consisting of Windows Server, Active Directory Domain Services (AD DS), Group Policy Objects (GPOs), Role-Based Access Control (RBAC), Windows Event Forwarding (WEF), Microsoft Defender, and Splunk Enterprise SIEM.

Throughout the project lifecycle, security events were generated, collected, forwarded, analyzed, and visualized to demonstrate practical Security Operations Centre (SOC) workflows. The implementation provides hands-on experience with identity management, endpoint monitoring, security event investigation, log management, and threat detection.

This project demonstrates skills commonly required for SOC Analyst, Security Engineer, Blue Team Analyst, and Cybersecurity Operations roles.

Project Scope:

This portfolio consists of ten interconnected security laboratories that simulate the deployment and monitoring of an enterprise Windows environment. The project covers identity management, access control, endpoint security, centralized event collection, SIEM deployment, and threat detection using industry-standard technologies.

Metric	Value
Total Labs Completed	10
Domain Controllers	1
Windows Endpoints	1
Splunk Server	1
Security Alerts Created	4
Security Reports Created	1+
Dashboard Panels Created	5
Event IDs Investigated	4624, 4625, 4720, 4728, 4732, 4740
SIEM Platform	Splunk Enterprise 9.4.3

Project Objectives

The primary objectives of this project were:

- Deploy an enterprise Active Directory environment.
- Configure Organizational Units (OUs), Users, and Security Groups.
- Implement DNS and Domain Services.
- Apply security controls using Group Policy Objects.

- Configure Role-Based Access Control (RBAC).
- Implement secure NTFS and Shared Folder permissions.
- Generate and investigate Windows security events.
- Configure Windows Event Forwarding (WEF).
- Validate Microsoft Defender security detections.
- Deploy Splunk Enterprise SIEM.
- Configure Universal Forwarders.
- Centralize Windows log collection.
- Build monitoring dashboards.
- Develop practical threat detection use cases.

Lab Environment

Infrastructure Components

Domain Controller

- Windows Server
- Active Directory Domain Services
- DNS Server
- Group Policy Management
- Organizational Unit Structure

Client Workstation

- Windows 10
- Domain Joined
- Security Event Generation
- Event Forwarding Client

Security Monitoring Server

- Ubuntu Server
- Splunk Enterprise
- Splunk Receiving Port (9997)

- Search & Reporting Application
- Dashboard Environment

Log Collection Components

- Splunk Universal Forwarder
- Windows Event Forwarding
- Windows Event Collector
- Microsoft Defender

LAB-01 – Active Directory Infrastructure

Objective

Deploy foundational Active Directory infrastructure.

Key Tasks

- Installed Active Directory Domain Services.
- Configured DNS services.
- Created Organizational Units.
- Created HR, IT, and Finance groups.
- Added users and group memberships.
- Validated domain functionality.

Result

Successfully established an enterprise Active Directory environment for centralized identity management.

LAB-02 – Domain Join & DNS Verification

Objective

Connect client systems to the domain and validate DNS functionality.

Key Tasks

- Configured client networking.

- Configured DNS resolution.
- Joined client computer to Active Directory domain.
- Validated authentication and domain membership.
- Verified DNS records.

Result

Successfully integrated client systems into the Active Directory environment.

LAB-03 – Group Policy Security Controls

Objective

Implement centralized security controls.

Key Tasks

- Password Policy Configuration.
- Account Lockout Policy.
- Control Panel Restrictions.
- USB Device Restrictions.
- User Security Controls.
- GPO Deployment Verification.

Result

Successfully enforced organization-wide security policies through Group Policy Objects.

LAB-04 – File Server & NTFS Permissions

Objective

Implement secure file sharing and RBAC.

Key Tasks

- Created departmental shared folders.
- Configured NTFS permissions.

- Configured Share permissions.
- Implemented Role-Based Access Control.
- Tested authorized and unauthorized access.
- Validated department-specific access rights.

Result

Successfully deployed secure file-sharing infrastructure with RBAC controls.

LAB-05 – Windows Event Monitoring & Investigation

Objective

Generate and investigate Windows Security Events.

Key Tasks

- Generated successful logon events.
- Generated failed logon events.
- Triggered account lockout events.
- Created user creation events.
- Investigated Event IDs 4624, 4625, and 4740.
- Validated security logging.

Result

Successfully identified and analyzed critical Windows security events used in SOC investigations.

LAB-06 – Microsoft Defender Security Monitoring

Objective

Validate endpoint threat detection capabilities.

Key Tasks

- Generated Defender security alerts.
- Performed EICAR malware simulation.

- Verified threat detection functionality.
- Monitored Windows Event Collector services.

Result

Successfully demonstrated endpoint protection and threat detection capabilities.

LAB-07 – Windows Event Forwarding (WEF)

Objective

Centralize Windows security event collection.

Key Tasks

- Configured WinRM using GPO.
- Configured WEF subscriptions.
- Configured Windows Event Collector.
- Added forwarding clients.
- Applied subscription filters.
- Validated event forwarding.

Result

Successfully centralized security event collection across the environment.

LAB-08 – Splunk SIEM Deployment

Objective

Deploy and configure Splunk Enterprise SIEM.

Key Tasks

- Installed Ubuntu Server.
- Installed Splunk Enterprise.
- Configured network connectivity.
- Enabled Splunk services.
- Enabled receiving port 9997.

- Installed Universal Forwarder.
- Connected Windows logs to Splunk.
- Verified log ingestion.
- Created monitoring dashboards.
- Conducted log analysis and event investigations.

Result

Successfully deployed a fully operational SIEM platform for centralized log analysis and monitoring.

LAB-09 – Threat Detection & Security Use Cases

Objective

Develop practical security monitoring and threat detection use cases using Splunk Enterprise and Windows Security Event Logs.

Key Tasks

• Verified Windows Security Log ingestion into Splunk.
• Detected failed logon attempts (Event ID 4625).
• Monitored successful logons (Event ID 4624).
• Detected new user account creation (Event ID 4720).
• Detected group membership changes (Event IDs 4728 & 4732).
• Created custom SPL searches for security monitoring.
• Built a centralized Threat Detection Dashboard.
• Validated detection use cases using real Windows security events.

Result

Successfully developed and validated multiple threat detection use cases using Windows Security Event Logs and Splunk Enterprise. Security monitoring dashboards were created to identify authentication events, account creation activity, account lockouts, and group membership changes within an Active Directory environment.

LAB-10 – Detection Engineering & Alerting

Objective

Implement automated threat detection, alerting, reporting, and monitoring capabilities using Splunk Enterprise to support proactive security operations and incident response workflows.

Key Tasks

• Created failed login detection use cases using Windows Security Event ID 4625.
• Created account lockout monitoring alerts using Event ID 4740.
• Created new user account creation alerts using Event ID 4720.
• Created group membership change alerts using Event IDs 4728 and 4732.
• Configured scheduled alert execution and trigger conditions.
• Created security reports for authentication-related events.
• Developed alert monitoring visualizations within the Threat Detection Dashboard.
• Validated detections using Windows Security Event Logs collected from domain-joined systems.

Result

Successfully implemented automated security monitoring and alerting capabilities within Splunk Enterprise, enabling continuous detection of authentication anomalies, account management activities, and privilege-related changes across the environment.

Overall Project Achievements

- Built a complete enterprise Active Directory environment.
- Deployed and managed Domain Controller services.
- Implemented DNS and Active Directory integration.
- Applied Group Policy security controls.
- Configured Role-Based Access Control (RBAC).
- Managed user, group, and organizational unit administration.
- Implemented Microsoft Defender security monitoring.
- Configured centralized Windows Event Forwarding (WEF).
- Deployed Splunk Enterprise SIEM.
- Integrated Splunk Universal Forwarders.
- Centralized Windows Security Event collection.
- Developed custom SPL searches and reports.
- Created security monitoring dashboards.
- Implemented threat detection use cases.
- Configured automated security alerts.

- Performed security event investigations and log analysis.
- Demonstrated practical SOC and Blue Team workflows.

Skills Demonstrated

Identity & Access Management

- Active Directory Administration
- User Management
- Group Management
- Organizational Unit Design

Windows Administration

- DNS Management
- Domain Administration
- Group Policy Management
- RBAC Implementation

Security Operations

- Event Analysis
- Security Monitoring
- Threat Detection
- Log Investigation

Endpoint Security

- Microsoft Defender
- EICAR Testing
- Security Event Validation

SIEM Technologies

- Splunk Enterprise
- Universal Forwarder
- Search Processing Language (SPL)

- Dashboard Development
- Security Analytics

Technologies Used

- Windows Server
- Windows 10
- Active Directory Domain Services
- DNS
- Group Policy
- Microsoft Defender
- Windows Event Forwarding
- Windows Event Collector
- Ubuntu Server
- Splunk Enterprise
- Splunk Universal Forwarder

Final Conclusion

This project demonstrates the successful design, deployment, administration, monitoring, and security analysis of a complete enterprise Windows environment integrated with Splunk Enterprise SIEM.

The project covers the full lifecycle of enterprise security operations, including Active Directory administration, identity and access management, Group Policy implementation, RBAC configuration, endpoint security monitoring, centralized log collection, threat detection, alerting, reporting, and security investigations.

Through ten hands-on labs, practical experience was gained in technologies and workflows commonly used by Security Engineers, SOC Analysts, Blue Team Analysts, System Administrators, and Detection Engineers.

The completed environment reflects real-world enterprise infrastructure and demonstrates the ability to build, secure, monitor, and investigate activities across a centralized security monitoring platform using industry-standard tools and methodologies.